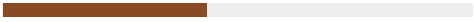
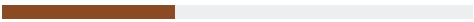


CompTIA A+

Core 1 (220-1201) · Core 2 (220-1202)

EXAM DOMAINS		WEIGHT
CORE 1 · 220-1201		
Mobile Devices		13%
Networking		23%
Hardware		25%
Virtualization and Cloud Computing		11%
Hardware and Network Troubleshooting		28%
CORE 2 · 220-1202		
Operating Systems		28%
Security		28%
Software Troubleshooting		23%
Operational Procedures		21%

Two exams, both required · up to 90 questions and 90 minutes each

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PART 1

Core 1

Exam 220-1201. Up to 90 questions, 90 minutes, passing score 675 on a scale of 100–900.

01

Mobile Devices

A+ CORE 1 · 220-1201 · DOMAIN 1 · 13% OF THE EXAM

This domain covers supporting laptops, tablets, and phones: swapping parts, connecting accessories, and setting up email, sync, and device management.

Laptop Hardware Components

Laptops use smaller, often proprietary parts. Before opening one, check the service manual, power it off, unplug it, and **disconnect the battery first**. Use ESD protection and keep track of screws (they're often different lengths).

COMPONENT	WHAT TO KNOW
Battery	Usually lithium-ion. Many modern laptops have internal batteries that need disassembly. A swollen battery is a safety hazard : stop using it, don't puncture it, and dispose of it properly.
Keyboard / keys	Often replaced as a whole assembly. Some are part of the top case (palm rest).
RAM	SODIMM (Small Outline DIMM), smaller than desktop DIMMs. Many thin laptops have RAM soldered to the board (not upgradeable).
Storage	2.5-inch SATA drives (HDD or SSD), M.2 SSDs (SATA or NVMe), or older mSATA. When migrating, clone the old drive or do a fresh install.
Wireless card	Usually M.2. Wi-Fi antenna wires run up into the display lid , so reconnect them carefully during screen or card replacement.
Camera / webcam / microphone	Located in the display bezel. Check privacy shutters and OS privacy settings before assuming hardware failure.
Biometrics	Fingerprint readers and IR cameras for facial recognition (Windows Hello).
NFC	Near Field Communication, for tap-to-pay and pairing. Very short range (~4 cm).

Mobile Display Components

- **LCD types**: IPS (best color and viewing angles), TN (fast response, poor angles, cheap), VA (good contrast, between the two). LCDs need a backlight, now LED.
- **OLED**: each pixel makes its own light, so blacks are true black and contrast is excellent. Can suffer **burn-in**.

- **Mini-LED:** an LCD with thousands of tiny backlight zones for better contrast.
- **Digitizer:** the touch-sensitive layer on top of the display. If touch is broken but the picture is fine, suspect the digitizer.
- **Inverter:** only in old laptops with CCFL backlights. A dim screen that's still faintly visible with a flashlight means a backlight or inverter failure.

Connecting and Configuring Accessories

CONNECTION METHODS

METHOD	NOTES
USB / USB-C	USB-C is reversible and carries data, video, and power (USB Power Delivery). Most Android phones and modern iPhones use it.
Lightning	Apple's older proprietary connector for iPhones and iPads.
Serial	Legacy RS-232. Still used for console cables to network gear.
Bluetooth	Wireless PAN, about 10 m. Pairing: enable Bluetooth → make the device discoverable → select it → enter or confirm the PIN → test.
NFC	Tap-to-connect and mobile payments.
Hotspot	The phone shares its cellular data connection over Wi-Fi.
Tethering	Sharing the phone's connection over USB (or Bluetooth) to one device.

ACCESSORIES

- **Docking station:** adds ports *and* often extra capabilities (extra displays, charging, sometimes drive bays). Usually connects through USB-C/Thunderbolt or a proprietary connector.
- **Port replicator:** simpler. Just duplicates the laptop's existing ports for convenient one-cable connection.
- Other accessories: touch pens and styluses, headsets, speakers, webcams, trackpads and drawing pads.

Mobile Device Network Connectivity and Application Support

- **Cellular data:** 4G LTE and 5G. Can be turned off to avoid data charges. An **eSIM** is a programmable SIM built into the phone.
- **Airplane mode** turns off all radios. Wi-Fi and Bluetooth can be re-enabled individually.
- **Location services:** GPS (satellite-based, works anywhere outdoors) and cellular location (triangulation from towers).
- **MDM (Mobile Device Management):** centrally enforces policies on devices: passcodes, encryption, approved apps, remote wipe. **MAM (Mobile Application Management)** manages only the corporate apps and data, which suits BYOD.

- **BYOD vs. corporate-owned:** BYOD devices belong to the user, so policies usually separate work data (containers, MAM) instead of controlling the whole device.
- **Two-factor authentication** and **authenticator apps** (TOTP codes).
- **Synchronization:** keeps mail, calendar, contacts, photos, and files in sync across devices. Can sync to the **cloud** (iCloud, Google, Microsoft 365), to a **desktop** (via USB or Wi-Fi), or through **Microsoft Exchange / ActiveSync** for corporate mail.

EMAIL CONFIGURATION

PROTOCOL	PORT (SECURE)	NOTES
POP3	110 (995)	Downloads mail to one device, typically deleting it from the server. Poor for multiple devices.
IMAP	143 (993)	Keeps mail on the server and syncs across all devices. Preferred today.
SMTP	25 / 587 (465)	Sends mail.
Exchange / ActiveSync	443	Microsoft corporate mail, calendar, contacts.

02

Networking

A+ CORE 1 · 220-1201 · DOMAIN 2 · 23% OF THE EXAM

Ports, addressing, Wi-Fi standards, and the hardware and tools used to build small networks.

TCP and UDP Ports and Protocols

PORT(S)	PROTOCOL	PURPOSE
20, 21	FTP	File transfer (unencrypted)
22	SSH	Secure remote command line
23	Telnet	Unencrypted remote command line
25	SMTP	Sending email
53	DNS	Name resolution
67, 68	DHCP	Automatic IP addressing
80	HTTP	Web (unencrypted)
110	POP3	Retrieving email
137–139	NetBIOS / NetBT	Legacy Windows name and file sharing
143	IMAP	Retrieving email (syncd)
161, 162	SNMP	Network device monitoring
389	LDAP	Directory services
443	HTTPS	Secure web
445	SMB / CIFS	Windows file and printer sharing
3389	RDP	Remote Desktop

- **TCP** is connection-oriented: it uses a three-way handshake and guarantees delivery (web, email, file transfer).
- **UDP** is connectionless: faster, no guarantee (streaming, VoIP, DNS lookups, DHCP).

Wireless Networking Technologies

STANDARD	NAME	BAND	MAX SPEED
802.11a	–	5 GHz	54 Mbps
802.11b	–	2.4 GHz	11 Mbps
802.11g	–	2.4 GHz	54 Mbps
802.11n	Wi-Fi 4	2.4 / 5 GHz	600 Mbps
802.11ac	Wi-Fi 5	5 GHz	~6.9 Gbps
802.11ax	Wi-Fi 6 / 6E	2.4 / 5 / 6 GHz	~9.6 Gbps
802.11be	Wi-Fi 7	2.4 / 5 / 6 GHz	~46 Gbps

- **2.4 GHz** has longer range but more interference, and only three non-overlapping channels (**1**, **6**, **11**). **5 GHz** is faster with more channels but shorter range. **6 GHz** is the cleanest and shortest range.
- **Channel width** (20/40/80/160 MHz): wider = faster but more prone to interference.
- **Bluetooth**: short-range PAN for peripherals. **NFC**: a few centimeters, for payments and pairing. **RFID**: radio tags for inventory, badges, and asset tracking.

Networked Hosts and Services

SERVER ROLE	WHAT IT DOES
DNS server	Resolves names to IP addresses.
DHCP server	Hands out IP configuration automatically.
File share	Central file storage (SMB on Windows, NFS on Linux).
Print server	Manages print queues and drivers for shared printers.
Mail server	Sends, receives, and stores email.
Syslog server	Collects logs from devices.
Web server	Hosts websites (HTTP/HTTPS).
AAA server	Authentication, Authorization, Accounting (e.g., RADIUS, TACACS+).
Database server	Stores structured data (SQL).
NTP server	Keeps clocks synchronized.

INTERNET APPLIANCES AND OTHER SYSTEMS

- **Spam gateway:** filters junk and malicious mail before it reaches users.
- **UTM (Unified Threat Management):** one appliance combining firewall, antivirus, IPS, content filtering, and VPN.
- **Load balancer:** spreads traffic across multiple servers.
- **Proxy server:** makes web requests on behalf of clients for filtering, caching, and privacy.
- **Legacy and embedded systems:** e.g., SCADA (industrial control). Often can't be patched, so isolate them on their own network.
- **IoT devices:** smart thermostats, cameras, speakers, lights. Put them on a separate network segment.

Network Configuration Concepts

DNS

RECORD	PURPOSE
A / AAAA	Name → IPv4 / IPv6 address
CNAME	Alias for another name
MX	Mail server for the domain
TXT	Text records, used for email security below

- **SPF (Sender Policy Framework):** a TXT record listing which servers are allowed to send mail for your domain.
- **DKIM (DomainKeys Identified Mail):** digitally signs outgoing mail so receivers can verify it wasn't altered.
- **DMARC:** tells receiving servers what to do when SPF or DKIM checks fail (none, quarantine, reject) and sends reports.

REMEMBER

These three work together to fight spam and email spoofing. SPF = who can send, DKIM = signature, DMARC = the policy.

DHCP

- **Scope:** the range of addresses to hand out. **Exclusions:** addresses in the scope that won't be handed out. **Reservations:** a fixed IP for a specific MAC address. **Leases:** how long a client keeps its address.

VLANS AND VPNS

- **VLAN:** splits one physical switch into separate logical networks (e.g., employees, guests, phones).
- **VPN:** an encrypted tunnel over the internet, connecting remote users or sites to the private network.

SOHO Networking: IP Addressing

- **IPv4:** 32-bit (e.g., 192.168.1.10). **IPv6:** 128-bit hex (e.g., 2001:db8::1).
- **Private ranges** (not routable on the internet): **10.0.0.0/8**, **172.16.0.0 – 172.31.255.255**, **192.168.0.0/16**. Everything else is public (apart from special ranges).
- **APIPA:** **169.254.x.x**. Self-assigned when DHCP is unreachable. Means the device can only talk to other APIPA hosts on the same segment.
- **Static** addressing is set by hand (servers, printers). **Dynamic** comes from DHCP (most clients).
- **Subnet mask:** defines which part of the address is the network (e.g., 255.255.255.0 = /24).
- **Default gateway:** the router address used to reach other networks and the internet. Wrong gateway = local access works, internet doesn't.
- **Loopback:** 127.0.0.1 (IPv4) or ::1 (IPv6) tests the local TCP/IP stack.

Internet Connection Types and Network Types

CONNECTION	NOTES
Fiber	Fastest and most reliable. Light over glass. Uses an ONT (Optical Network Terminal) at the home.
Cable	Coax shared with TV service. Uses a cable modem (DOCSIS). Bandwidth shared with neighbors.
DSL	Runs over phone lines. Speed drops with distance from the provider's office.
Satellite	Available almost anywhere. Traditional satellite has high latency; LEO satellite (e.g., Starlink) is much lower.
Cellular	4G/5G, via phone hotspot or a dedicated cellular router.
WISP	Wireless Internet Service Provider: fixed wireless antennas, common in rural areas.

Network types by size: PAN (personal) → LAN (building) → WLAN (wireless LAN) → MAN (city) → WAN (wide area). A **SAN** is a storage area network.

Networking Hardware

DEVICE	PURPOSE
Router	Connects different networks and forwards traffic between them (e.g., your LAN to the internet).
Unmanaged switch	Plug-and-play. No configuration.
Managed switch	Configurable: VLANs, port security, monitoring, QoS.
Access point	Connects wireless devices to the wired network.
Patch panel	Where in-wall cable runs terminate in a wiring closet.
Firewall	Filters traffic based on rules.
PoE / PoE injector	Power over Ethernet: sends power with data to APs, cameras, phones. An injector adds PoE when the switch doesn't support it.
Hub	Legacy. Repeats everything to every port (one collision domain).
Cable / DSL modem	Converts the provider's signal to Ethernet.
ONT	Optical Network Terminal: the fiber equivalent of a modem.
NIC	Network Interface Card: the device's network adapter.
SDN	Software-Defined Networking: network controlled centrally by software.

Networking Tools

TOOL	USE
Crimper	Attaches RJ45 connectors to the end of a cable.
Cable stripper	Removes the outer jacket without damaging the wires.
Punchdown tool	Seats wires into a patch panel or keystone jack.
Cable tester	Checks continuity and correct wiring (wire map).
Toner probe	Traces a cable to find its other end.
Loopback plug	Tests whether a network port or NIC works by looping its signal back.
Wi-Fi analyzer	Shows wireless networks, channels, and signal strength. Finds interference.
Network tap	Copies traffic from a link for monitoring.

REMEMBER

Copper cable wiring: **T568B** order is white/orange, orange, white/green, blue, white/blue, green, white/brown, brown. Same standard on both ends = straight-through.

03

Hardware

A+ CORE 1 · 220-1201 · DOMAIN 3 · 25% OF THE EXAM

Know what each component does, how to tell types apart, and how to pick compatible parts. Expect drag-and-drop questions matching connectors and parts.

Cables and Connectors

NETWORK CABLES

- **Copper twisted pair:** Cat 5 (100 Mbps), Cat 5e (1 Gbps), Cat 6 (1 Gbps, 10 Gbps up to 55 m), Cat 6a (10 Gbps), Cat 8 (25/40 Gbps, short runs). Max length **100 m**. **Plenum** cable is required in air-handling spaces (fire resistant, less toxic smoke). **STP** (shielded) resists interference; **UTP** is standard.
- **Coaxial:** RG-6, used for cable internet and TV, with F-type connectors.
- **Fiber:** **single-mode** (laser, long distance) and **multimode** (LED, shorter distance). Connectors: **LC** (small, latching), **SC** (square push-pull), **ST** (round twist-lock).

PERIPHERAL AND VIDEO CABLES

CABLE / CONNECTOR	NOTES
USB 2.0	480 Mbps
USB 3.x	5 Gbps (3.2 Gen 1), 10 Gbps (Gen 2), 20 Gbps (Gen 2x2). Ports are often blue.
USB4 / USB-C	USB-C is the reversible <i>connector</i> . USB4 runs up to 40 Gbps (USB4 v2 up to 80) and can carry video and power.
Thunderbolt	Intel/Apple standard over USB-C (Thunderbolt 3/4 = 40 Gbps). Carries data, video, and power, and supports daisy-chaining.
Lightning	Apple's older mobile connector.
Serial (DB-9)	Legacy. Console connections.
HDMI	Digital video and audio. Most common for TVs and monitors.
DisplayPort	Digital video and audio. Common on PCs and monitors. Supports daisy-chaining (MST).
DVI	Digital (DVI-D), analog (DVI-A), or both (DVI-I). Video only.
VGA	Legacy analog video. Blue 15-pin DE-15 connector.

CABLE / CONNECTOR	NOTES
SATA	Internal drive data cable (7-pin). eSATA is the external version.
Molex	Old 4-pin power connector for drives and fans.

REMEMBER

Only HDMI and DisplayPort carry audio among the video connectors. VGA is the only analog-only one.

RAM

TYPE	PINS (DIMM / SODIMM)	NOTES
DDR3	240 / 204	Legacy
DDR4	288 / 260	Common in many current systems
DDR5	288 / 262	Current. Higher speeds, on-die ECC, two subchannels per module.

- RAM generations are **not interchangeable**. The notch is in a different spot, so a DDR4 stick physically won't fit a DDR5 slot.
- **SODIMM**: laptop form factor. **DIMM**: desktop form factor.
- **ECC (Error-Correcting Code)**: detects and fixes single-bit memory errors. Used in servers. Requires motherboard and CPU support. Non-ECC is standard in consumer PCs.
- **Channels**: single, dual, triple, quad. Install matching modules in the correct slots (check the manual, often same-colored slots) for multi-channel bandwidth.
- **Virtual RAM / page file**: disk space used as overflow memory when physical RAM runs out. Much slower than real RAM.

Storage Devices

TYPE	DETAILS
HDD	Spinning platters. Speeds: 5,400 / 7,200 / 10,000 / 15,000 RPM (higher = faster). Form factors: 3.5" (desktop) and 2.5" (laptop).
SATA SSD	No moving parts, much faster than HDD. Limited by SATA III's ~600 MB/s.
NVMe SSD	Connects over PCIe directly. Several times faster than SATA SSDs.
M.2	A <i>form factor</i> , not a speed. An M.2 drive can be SATA or NVMe; check what the slot supports. Common sizes: 2242, 2260, 2280 (22 mm wide × 80 mm long).
mSATA	Older small SATA SSD form factor.
Removable	USB flash drives, memory cards (SD, microSD), optical drives (CD, DVD, Blu-ray).

RAID

LEVEL	HOW IT WORKS	MIN. DRIVES	FAULT TOLERANCE
RAID 0	Striping across drives. Fast, full capacity.	2	None. One drive fails = all data lost.
RAID 1	Mirroring; identical copies.	2	Survives one drive failure. 50% capacity.
RAID 5	Striping with distributed parity.	3	Survives one drive failure.
RAID 6	Striping with double parity.	4	Survives two drive failures.
RAID 10 (1+0)	Mirrored pairs, then striped.	4	Survives one failure per mirror. Fast.

REMEMBER

RAID is not a backup. It protects against drive failure, not deletion, ransomware, or fire.

Motherboards, CPUs, and Add-On Cards

MOTHERBOARD FORM FACTORS

FORM FACTOR	SIZE	NOTES
ATX	12 × 9.6 in	Standard desktop. Most expansion slots.
microATX (mATX)	9.6 × 9.6 in	Smaller, fewer slots. Fits most ATX cases.
Mini-ITX	6.7 × 6.7 in	Small form factor PCs. Usually one expansion slot.

MOTHERBOARD FEATURES

- **Expansion slots:** PCIe in x1, x4, x8, and x16 sizes. Graphics cards use x16. (PCI is legacy.)
- **Connectors and headers:** SATA, M.2, 24-pin main power, 4/8-pin CPU power, front-panel headers (power button, LEDs, USB, audio), fan headers.
- **CPU socket:** must match the CPU. Intel uses **LGA** (pins on the motherboard). AMD's desktop AM5 is also LGA; older AMD sockets (AM4) used **PGA** (pins on the CPU).
- **Multisocket:** server boards can hold multiple CPUs.
- **Server vs. desktop vs. mobile** boards differ in size, ECC support, and expandability.

BIOS / UEFI SETTINGS

- **UEFI** replaced legacy BIOS. It supports GPT drives over 2 TB, a mouse-driven interface, faster boot, and **Secure Boot**.
- Common settings: **boot order**, **USB permissions** (disable ports for security), **fan and temperature** settings, **virtualization support** (Intel VT-x / AMD-V, needed for VMs), **Secure Boot** (only runs signed bootloaders, blocks rootkits), and **boot / setup passwords**.
- **TPM (Trusted Platform Module):** a chip (or firmware) that securely stores encryption keys. Used by BitLocker and required by Windows 11 (TPM 2.0).
- **HSM (Hardware Security Module):** a dedicated device for managing and protecting cryptographic keys, usually in enterprise environments.
- **CMOS battery:** a coin cell (CR2032) that keeps the clock and settings. When it dies, the system loses the date and time.

CPU FEATURES

- **Architectures:** **x64/x86** (Intel, AMD desktops and laptops) and **ARM** (phones, tablets, Apple Silicon Macs; very power efficient).
- **Cores and threads:** multicore CPUs run tasks in parallel. **Hyperthreading / SMT** lets each core run two threads.
- **Virtualization support** must be enabled in UEFI for hypervisors.

EXPANSION CARDS AND COOLING

- **Expansion cards:** sound card, video (graphics) card, capture card (records video input), NIC.
- **Cooling:** fans, **heat sinks**, **thermal paste** or thermal pads (fill microscopic gaps between CPU and heat sink, so reapply when reinstalling), and **liquid cooling** (pump, radiator, tubes).

Power Supplies

- **Input:** 110–120 VAC (North America) or 220–240 VAC (Europe and elsewhere). Most PSUs auto-switch. Older ones have a voltage switch; setting it wrong can destroy the PSU.
- **Output:** **+3.3 V**, **+5 V**, and **+12 V DC**. The 12 V rail powers the CPU, GPU, and drives.
- **Connectors:** 24-pin motherboard, 4/8-pin CPU, 6/8-pin PCIe (graphics), 12VHPWR / 12V-2x6 (newest high-power GPUs), SATA power.

- **Wattage rating:** add up component draw and leave headroom. High-end GPUs need large PSUs.
- **Modular** PSUs let you attach only the cables you need (better airflow). **Redundant** PSUs (servers) keep running if one unit fails.
- **80 PLUS** ratings indicate efficiency (Bronze, Silver, Gold, Platinum, Titanium).

Displays

- **Panel types:** IPS (best color and angles), TN (fastest response, worst angles), VA (high contrast), OLED (true blacks, risk of burn-in), mini-LED.
- **Resolution:** number of pixels, e.g., 1920×1080 (1080p), 2560×1440 (1440p), 3840×2160 (4K).
- **Refresh rate:** times per second the image redraws (Hz). 60 Hz standard, 144 Hz+ for gaming.
- **Aspect ratio:** 16:9 most common; 16:10 and 21:9 (ultrawide) also exist.
- **Color gamut:** the range of colors a display can show (sRGB, DCI-P3). Matters for design work.
- **Touch screens** and **digitizers** add touch input.

Printers and Multifunction Devices

DEPLOYMENT AND CONFIGURATION

- **Drivers:** **PCL** (fast, common for office printing) and **PostScript** (precise, device-independent, preferred for graphics). Wrong driver = garbled output.
- **Firmware** updates fix bugs and security holes.
- **Connectivity:** USB, Ethernet, Wi-Fi. Shared via a **print server** or directly from a client (public/shared devices).
- **Configuration:** duplex (two-sided), orientation, tray selection, quality.
- **Security:** user authentication, **badging** (release jobs with an ID card), audit logs, **secured prints** (job held until the user enters a PIN at the printer).
- **Network scan services:** scan to email, to SMB folder, or to cloud. **ADF** (Automatic Document Feeder) vs. flatbed scanner.

LASER PRINTERS: THE SEVEN-STEP IMAGING PROCESS

STEP	WHAT HAPPENS
1. Processing	The printer builds the page image in memory.
2. Charging	The primary charge roller (or corona wire) applies a uniform negative charge to the imaging drum.
3. Exposing	The laser draws the image, neutralizing the charge where toner should stick.
4. Developing	Toner is attracted to the exposed areas of the drum.
5. Transferring	The transfer roller (or belt, for color) pulls toner from the drum onto the paper.
6. Fusing	The fuser's heat and pressure melt toner onto the paper.
7. Cleaning	A blade scrapes leftover toner off the drum, and the charge is removed.

REMEMBER

Mnemonic: **People Can't Ever Develop Their Favorite Colors**. Toner that smudges or wipes off = **fuser** problem.

- **Maintenance:** replace toner, apply a **maintenance kit** (new fuser, rollers) at the recommended page count, calibrate, and clean with a toner-safe vacuum (regular vacuums can be damaged by or ignite toner).

OTHER PRINTER TYPES

TYPE	HOW IT WORKS	MAINTENANCE
Inkjet	Sprays ink from cartridges through a print head. Parts: cartridge, print head, roller, feeder, duplexing assembly, carriage and belt.	Clean heads, nozzle check, calibrate (align), replace cartridges, clear jams.
Thermal	Heats special heat-sensitive paper (receipts, labels). Parts: feed assembly, heating element.	Replace paper, clean the heating element (isopropyl alcohol), remove debris. Keep paper away from heat.
Impact	Pins strike an inked ribbon against paper. Used for multipart forms (carbon copies). Uses tractor feed.	Replace the ribbon, print head, and paper.
3D	Builds objects layer by layer from filament (FDM) or liquid resin (SLA).	Level the print bed, keep the nozzle clean, store filament dry.

04

Virtualization and Cloud Computing

A+ CORE 1 · 220-1201 · DOMAIN 4 · 11% OF THE EXAM

Virtualization Concepts

- **Virtual machine (VM):** a complete computer running in software on a host. Each VM has its own OS.
- **Hypervisor:** the software that creates and runs VMs. **Type 1 (bare metal)** runs directly on hardware: VMware ESXi, Microsoft Hyper-V, Proxmox, KVM. Used in data centers. **Type 2 (hosted)** runs as an app on a normal OS: VirtualBox, VMware Workstation. Used on desktops.
- **Why use VMs:** **sandboxing** (safely test suspicious files), **test development**, running **legacy software or OSs**, and **cross-platform** needs (run Linux on Windows).
- **Containers:** package an app with its dependencies but share the host's OS kernel. Much lighter and faster to start than VMs (e.g., Docker).
- **VDI (Virtual Desktop Infrastructure):** desktops run on central servers and users access them remotely.
- **Application virtualization:** apps run in an isolated environment or stream from a server without a full install.
- **Requirements:** enough CPU, RAM, and storage for host *plus* guests, virtualization enabled in UEFI, and network configuration (NAT, bridged, or internal virtual switches). Each VM needs patching and security like a physical machine.

REMEMBER

Type 1 = bare metal (enterprise). Type 2 = runs as an application on a desktop OS.

Cloud Computing Concepts

DEPLOYMENT MODELS

- **Public:** shared infrastructure owned by a provider (AWS, Azure, Google Cloud).
- **Private:** dedicated to one organization.
- **Hybrid:** a mix of public and private.
- **Community:** shared by several organizations with common needs (e.g., government agencies, hospitals).

SERVICE MODELS

MODEL	YOU GET	EXAMPLE
SaaS	A finished application. You just use it.	Microsoft 365, Gmail, Salesforce
PaaS	A platform to build and run your own apps without managing servers.	Azure App Service, Heroku
IaaS	Raw infrastructure: VMs, storage, networking. You manage the OS and up.	AWS EC2, Azure VMs

CLOUD CHARACTERISTICS

- **Shared vs. dedicated resources:** shared hardware is cheaper; dedicated gives isolation.
- **Metered utilization:** you pay for what you use, like a utility bill.
- **Elasticity:** resources automatically scale up and down with demand.
- **Availability:** redundancy across data centers keeps services online.
- **File synchronization:** files stay in sync across devices (OneDrive, Google Drive, Dropbox).
- **Multitenancy:** many customers share the same infrastructure, isolated from one another.

05

Hardware and Network Troubleshooting

A+ CORE 1 · 220-1201 · DOMAIN 5 · 28% OF THE EXAM

Most questions here describe a symptom and ask for the most likely cause or the best next step. Learn to match symptoms to causes.

The Troubleshooting Methodology

STEP	DETAILS
1. Identify the problem	Gather information from the user, identify user changes, perform backups before making changes, inquire about environmental or infrastructure changes.
2. Establish a theory of probable cause	Question the obvious. If necessary, research using internal or external sources.
3. Test the theory	If confirmed, determine next steps to resolve. If not, re-establish a new theory or escalate.
4. Establish a plan of action and implement the solution	Refer to vendor instructions for guidance.
5. Verify full system functionality	Implement preventive measures if applicable.
6. Document	Findings, actions, and outcomes.

REMEMBER

Note the A+ detail: **back up data before making changes** is part of step 1. “Document” is always last.

Motherboards, RAM, CPU, and Power

SYMPTOM	LIKELY CAUSE / FIX
No power at all	Power cable, outlet, PSU switch, failed PSU. Test with a known-good PSU or a PSU tester.

SYMPTOM	LIKELY CAUSE / FIX
POST beep codes	Hardware failure found during startup. Look up the pattern in the motherboard manual (often RAM or video).
Black screen, fans spin	Reseat RAM and the graphics card, check the monitor input and cable.
Blue screen (BSOD) / macOS spinning pinwheel / kernel panic	Drivers, faulty RAM, overheating. Check the stop code; run memory diagnostics.
Intermittent shutdowns	Overheating (dust, failed fan, dried thermal paste) or a failing PSU.
Burning smell or smoke	Power off immediately and unplug. Usually a failed PSU or component.
Swollen or leaking capacitors	Failing motherboard or PSU. Replace.
Grinding noise	Failing fan or HDD.
Sluggish performance	Overheating (CPU throttling), not enough RAM, failing drive, malware.
Wrong date and time on every boot	Dead CMOS battery. Replace the coin cell.
Random app crashes, unstable system	Bad RAM, overheating, power issues.

Storage Drives and RAID

SYMPTOM	LIKELY CAUSE / FIX
Clicking or grinding noises	HDD mechanical failure. Back up immediately and replace.
“Bootable device not found”	Wrong boot order, loose cable, failed drive, corrupted boot sector.
S.M.A.R.T. failure warning	The drive’s self-monitoring predicts failure. Back up and replace.
Slow read/write times, low IOPS	Failing drive, nearly full drive, HDD fragmentation.
Data loss or corruption	Failing drive, bad sectors, sudden power loss. Run chkdsk; restore from backup.
RAID degraded or failed	A member drive failed. Replace it and let the array rebuild (if the RAID level allows).
LED status indicators	Drive and RAID controller LEDs signal activity or faults. Check the vendor’s meanings.

Displays and Projectors

SYMPTOM	LIKELY CAUSE / FIX
No image	Wrong input source selected, loose or bad cable, monitor off.
Fuzzy image	Not running at native resolution.
Dim image	Brightness setting, failing backlight. Projector: aging bulb.
Flickering / flashing	Loose cable, wrong refresh rate, failing backlight or inverter.
Dead pixels	Permanently black or stuck-color pixels. Usually requires panel replacement.
Burn-in	Ghost image from static content, mostly OLED. Use screen savers, vary content.
Incorrect color display	Bad cable or connector, wrong color profile, failing GPU.
Projector shuts off	Overheating: clean the filter, check the fan and ventilation.
Projector won't light	Burned-out bulb.
No audio	Wrong output device selected, muted, cable doesn't carry audio (DVI, VGA).

Mobile Devices

SYMPTOM	LIKELY CAUSE / FIX
Poor battery health / short battery life	Aging battery, too many background apps, high screen brightness. Replace the battery if it's degraded.
Swollen battery	Stop using the device. Remove the battery safely and replace it. Fire hazard.
Won't charge properly	Bad cable or charger, dirty or damaged port.
Overheating	Heavy apps, blocked vents, failing battery, malware.
Liquid damage	Power off right away, don't charge, check liquid contact indicators.
Broken screen / digitizer issues	Touch not responding = digitizer. Replace the screen assembly.
Cursor drift / wrong touch location	Recalibrate the touchscreen or trackpad, clean it, update drivers.
Poor or no connectivity	Airplane mode, Wi-Fi or Bluetooth off, antenna wiring (laptops), out of range.
Physically damaged ports	Replace the port or board.
Unexpected behavior, battery drain, data use	Possible malware. Scan, remove suspicious apps.

Printers

SYMPTOM	LIKELY CAUSE / FIX
Vertical lines down the page	Laser: scratched drum or dirty components. Inkjet: clogged nozzles (clean the heads).
Garbled print (random characters)	Wrong or corrupt driver. Reinstall the correct driver.
Toner not fused (smears, rubs off)	Faulty fuser . Replace it.
Paper jams	Wrong paper type, worn pickup rollers, debris in the paper path.
Faded print	Low toner or ink, toner-saving mode.
Paper not feeding / multipage misfeed	Worn rollers, humid or stuck-together paper, wrong tray settings.
Speckling	Loose toner inside the printer. Clean with a toner vacuum.
Double / echo (ghost) images	Drum not cleaned properly, or a failing drum or cleaning blade.
Incorrect colors (chroma)	Low cartridge, needs calibration.
Jobs stuck in queue	Restart the Print Spooler service, clear the queue, check printer connection.
Grinding noise	Gear or roller problem, stuck paper.
Finishing issues	Staple jams, hole punch jams. Clear the finisher.
Wrong orientation or paper size	Check driver settings and tray configuration.

Wired and Wireless Networks

SYMPTOM	LIKELY CAUSE / FIX
Intermittent wireless	Interference (microwaves, other networks), distance, channel overlap. Use a Wi-Fi analyzer; change channel or band.
Slow network speeds	Congestion, interference, duplex mismatch, old cabling, ISP issue.
Limited connectivity / APIPA (169.254.x.x)	Can't reach DHCP. Check cable, DHCP server, and VLAN.
Jitter / poor VoIP quality	Inconsistent delay. Configure QoS to prioritize voice.
High latency	Congestion, long route, satellite link.
Port flapping	A switch port rapidly going up and down. Bad cable, failing NIC or transceiver.
Authentication failures	Wrong password or security type, expired credentials, RADIUS issue.
External interference	EMI near cables; other devices on the same wireless frequencies.
Intermittent internet	ISP outage, failing modem or router, loose connections.

REMEMBER

Can reach local devices but not the internet → check the **default gateway**. Can ping an IP but not a website name → check **DNS**.

Practice Questions

32 QUESTIONS · ANSWER KEY FOLLOWS

- 01 A user wants to connect a laptop to two external monitors, Ethernet, and power using one cable. Which accessory is the BEST choice?**
- A Port replicator
 - B Docking station
 - C KVM switch
 - D USB hub
- 02 A laptop's Wi-Fi signal became very weak after the screen was replaced. What is the MOST likely cause?**
- A Bad wireless driver
 - B Disconnected antenna wires in the display lid
 - C Failing battery
 - D Wrong SSID
- 03 Which email protocol keeps messages on the server and syncs them across multiple devices?**
- A POP3
 - B SMTP
 - C IMAP
 - D SNMP
- 04 A company lets employees use personal phones but wants to control only the work apps and data. What should be used?**
- A Full-device MDM wipe policy
 - B Mobile Application Management (MAM)
 - C Disabling Bluetooth
 - D A VPN only
- 05 Which port does RDP use?**
- A 22
 - B 443
 - C 445
 - D 3389
- 06 Which port is used for Windows file sharing with SMB?**
- A 139
 - B 389
 - C 445
 - D 110

- 07 A PC shows the IP address 169.254.12.4. What does this indicate?**
- A The PC has a static IP
 - B The PC could not reach a DHCP server
 - C The PC is using IPv6
 - D The DNS server is down
- 08 Which DNS record helps prevent email spoofing by listing servers allowed to send mail for a domain?**
- A MX
 - B CNAME
 - C SPF (TXT)
 - D AAAA
- 09 A technician needs to attach an RJ45 connector to a new Ethernet cable. Which tool is needed?**
- A Punchdown tool
 - B Crimper
 - C Toner probe
 - D Loopback plug
- 10 A rural home has no cable or fiber and wants low-latency internet. Which option is MOST likely the best fit?**
- A Geostationary satellite
 - B WISP or LEO satellite
 - C Dial-up
 - D Token ring
- 11 Which device provides power to an IP camera over its Ethernet cable when the switch doesn't support PoE?**
- A Patch panel
 - B PoE injector
 - C ONT
 - D Hub
- 12 A user's PC loses the correct date and time every time it's unplugged. What should be replaced?**
- A Power supply
 - B CMOS battery
 - C RAM
 - D Hard drive

- 13 Which RAID level provides striping with NO fault tolerance?**
- A RAID 0
 - B RAID 1
 - C RAID 5
 - D RAID 10
- 14 What is the minimum number of drives for RAID 5?**
- A 2
 - B 3
 - C 4
 - D 5
- 15 An M.2 SSD is installed but performs at SATA speeds. What is the MOST likely reason?**
- A It's a SATA M.2 drive, or the slot only supports SATA
 - B RAID 0 is not enabled
 - C The CMOS battery is dead
 - D The drive needs defragmenting
- 16 Which feature must be enabled in UEFI before running a hypervisor such as Hyper-V?**
- A Secure Boot
 - B Virtualization support (VT-x / AMD-V)
 - C USB permissions
 - D Fast boot
- 17 Which video connector carries ONLY analog signals?**
- A HDMI
 - B DisplayPort
 - C VGA
 - D DVI-D
- 18 What are the standard DC output voltages of a desktop power supply?**
- A 1.5 V, 3 V, 9 V
 - B 3.3 V, 5 V, 12 V
 - C 5 V, 12 V, 24 V
 - D 110 V, 220 V
- 19 Toner on printed pages smears when touched. Which laser printer component has failed?**
- A Pickup roller
 - B Fuser
 - C Laser
 - D Transfer roller

- 20 Which step of the laser imaging process applies a uniform negative charge to the drum?**
- A Exposing
 - B Charging
 - C Developing
 - D Transferring
- 21 A store needs a printer for receipts that uses no ink or toner. Which type fits?**
- A Impact
 - B Inkjet
 - C Thermal
 - D Laser
- 22 Which hypervisor type runs directly on the hardware without a host operating system?**
- A Type 1
 - B Type 2
 - C Container
 - D Emulator
- 23 A company wants to run developers' apps on a provider's platform without managing servers or operating systems. Which model is this?**
- A IaaS
 - B PaaS
 - C SaaS
 - D On-premises
- 24 Which cloud characteristic automatically adds resources during peak demand and removes them afterward?**
- A Multitenancy
 - B Elasticity
 - C Metered utilization
 - D File synchronization
- 25 A technician has gathered information from the user. According to the A+ methodology, what should be done before making changes?**
- A Document the findings
 - B Perform backups
 - C Escalate
 - D Verify functionality

- 26 A PC randomly shuts down after running for about 20 minutes, and the case feels hot. What is the MOST likely cause?**
- A Bad RAM
 - B Overheating from dust or a failed fan
 - C Corrupted driver
 - D Wrong boot order
- 27 A hard drive makes clicking noises and S.M.A.R.T. reports errors. What should the technician do FIRST?**
- A Defragment the drive
 - B Back up the data
 - C Run disk cleanup
 - D Update the BIOS
- 28 A phone's back case is bulging and won't sit flat. What should the technician do?**
- A Keep charging it to calibrate
 - B Stop using it and safely replace the battery
 - C Update the OS
 - D Factory reset
- 29 A printer outputs pages of random symbols and characters. What is the MOST likely fix?**
- A Replace the fuser
 - B Reinstall the correct print driver
 - C Replace the toner
 - D Clean the rollers
- 30 Print jobs are stuck in the queue and won't print or delete. What should be restarted?**
- A The DHCP client
 - B The Print Spooler service
 - C Windows Update
 - D The router
- 31 VoIP calls sound choppy, and monitoring shows inconsistent packet delay. What is the BEST fix?**
- A Configure QoS to prioritize voice traffic
 - B Replace the phone handsets
 - C Change the SSID
 - D Disable DHCP

- 32** A projector shuts off by itself after a few minutes of use. What is the MOST likely cause?
- A Wrong input source
 - B Overheating from a clogged filter or failed fan
 - C Bad HDMI cable
 - D Wrong resolution

Answer Key

#	ANSWER	WHY
01	B	A docking station adds extra capabilities like multiple displays and charging through one connection. (Domain 1)
02	B	Laptop Wi-Fi antennas run through the display lid and are easy to leave disconnected during screen work. (Domain 1)
03	C	IMAP syncs mail across devices. POP3 downloads mail to one device. (Domain 1/2)
04	B	MAM manages corporate apps and data without controlling the whole personal device, which suits BYOD. (Domain 1)
05	D	RDP uses TCP 3389. (Domain 2)
06	C	Modern SMB uses TCP 445. 137-139 is legacy NetBIOS. (Domain 2)
07	B	169.254.x.x is APIPA, which is self-assigned when DHCP fails. (Domain 2)
08	C	SPF is published as a TXT record. (Domain 2)
09	B	A crimper attaches connectors. A punchdown tool seats wires in patch panels and keystone jacks. (Domain 2)
10	B	Fixed wireless (WISP) and LEO satellite reach rural areas with far lower latency than traditional satellite. (Domain 2)
11	B	A PoE injector adds power onto the Ethernet line. (Domain 2)
12	B	The CMOS coin-cell battery keeps the clock running when the PC is off. (Domain 3/5)
13	A	RAID 0 stripes for speed. One drive failure loses everything. (Domain 3)
14	B	RAID 5 needs at least three drives. (Domain 3)
15	A	M.2 is a form factor. Drives and slots can be SATA or NVMe (PCIe). (Domain 3)
16	B	Hardware virtualization must be turned on in firmware for hypervisors. (Domain 3/4)
17	C	VGA is analog-only. (Domain 3)
18	B	PSUs output +3.3, +5, and +12 VDC. (Domain 3)
19	B	The fuser melts toner onto the paper. Unfused toner rubs off. (Domain 3/5)
20	B	Charging comes before exposing. (Domain 3)
21	C	Thermal printers heat special paper. (Domain 3)
22	A	Type 1 is bare metal (ESXi, Hyper-V, Proxmox). (Domain 4)
23	B	PaaS provides the platform; you bring the app. (Domain 4)
24	B	Elasticity scales with demand. (Domain 4)

#	ANSWER	WHY
25	B	A+ step 1 includes performing backups before making changes. (Domain 5)
26	B	Intermittent shutdowns under load with heat point to cooling problems. (Domain 5)
27	B	The drive is failing. Protect the data first, then replace the drive. (Domain 5)
28	B	A swollen battery is a fire hazard. (Domain 5)
29	B	Garbled output points to a wrong or corrupt driver. (Domain 5)
30	B	Restarting the Print Spooler clears stuck jobs. (Domain 5)
31	A	That's jitter. QoS prioritizes voice. (Domain 5)
32	B	Projectors shut down to protect themselves from heat. (Domain 5)

PART 2

Core 2

Exam 220-1202. Up to 90 questions, 90 minutes, passing score 700 on a scale of 100–900.

01

Operating Systems

A+ CORE 2 · 220-1202 · DOMAIN 1 · 28% OF THE EXAM

This domain is mostly Windows, with a solid chunk of macOS and Linux. Know the tools by name *and* by their shortcut (e.g., devmgmt.msc), and know the commands.

Operating System Types and File Systems

- **Desktop OSs:** Windows, macOS, Linux, ChromeOS (browser-centric, cloud-based). **Mobile OSs:** iOS, iPadOS, Android.
- **Vendor life-cycle limitations:** once an OS reaches end of life, it stops getting security updates. Running it is a vulnerability.
- **Compatibility:** apps and file systems don't always work across OSs. Check before upgrading or migrating.

FILE SYSTEM	USED BY	KEY FACTS
NTFS	Windows	Default Windows file system. Permissions, encryption (EFS), compression, journaling, huge file sizes.
ReFS	Windows Server / Pro for Workstations	Resilient File System. Built for data integrity and large storage pools. Can't boot Windows from it.
FAT32	Everything	Universal compatibility, but 4 GB max file size and no permissions.
exFAT	Flash drives, SD cards	Cross-platform (Windows and macOS) with no 4 GB limit. Best choice for large files on removable media.
ext4	Linux	Default on most Linux distributions. Journaling.
XFS	Linux (e.g., RHEL)	High-performance journaling file system for large files and volumes.
APFS	macOS, iOS	Apple File System. Optimized for SSDs, built-in encryption and snapshots.

REMEMBER

If a user can't copy a 6 GB file to a USB drive, the drive is probably FAT32. Reformat it as exFAT (or NTFS for Windows-only use).

OS Installations and Upgrades

BOOT METHODS

USB drive, **network boot (PXE)**, solid-state/flash drive, internet-based installation, external or hot-swappable drive, and an internal hard drive partition (recovery partition).

INSTALLATION TYPES

TYPE	DESCRIPTION
Clean install	Wipes the drive and installs fresh. Best performance, but apps and data must be restored.
In-place upgrade	Installs over the existing OS, keeping apps, files, and settings.
Image deployment	Copies a prebuilt, configured image to many machines at once.
Remote network installation	Installs over the network from a deployment server.
Zero-touch deployment	Devices configure themselves automatically when the user first signs in (e.g., Windows Autopilot).
Recovery partition	Hidden partition with the factory image, used to restore the PC.
Repair installation	Reinstalls Windows system files while keeping data and apps. Fixes corrupted OS files.
Third-party drivers	Some storage controllers need drivers loaded during setup or the installer won't see the disk.

PARTITIONING AND FORMATTING

	MBR	GPT
Max disk size	2 TB	Huge (about 9.4 ZB)
Max partitions	4 primary (or 3 + extended)	128 in Windows
Firmware	Legacy BIOS	UEFI (required for Secure Boot and Windows 11)

- **Quick format** just resets the file table. **Full format** also scans for bad sectors.
- **Before upgrading:** back up files and user preferences, check hardware compatibility, check app and driver support, and plan for feature updates.

Windows 10 and Windows 11 Editions

FEATURE	HOME	PRO	PRO FOR WORKSTATIONS	ENTERPRISE
Join a domain	No	Yes	Yes	Yes
Group Policy Editor (gpedit.msc)	No	Yes	Yes	Yes
BitLocker	No (device encryption only)	Yes	Yes	Yes
Host Remote Desktop	No (client only)	Yes	Yes	Yes
Max RAM (Win 11)	128 GB	2 TB	6 TB	6 TB
Buy at retail	Yes	Yes	Yes	No (volume licensing)

- **Workgroup:** peer-to-peer, each PC manages its own accounts (homes, tiny offices). **Domain:** centrally managed with Active Directory.
- **Windows 11 requirements:** 64-bit 1 GHz dual-core CPU (supported list), **4 GB RAM, 64 GB storage, UEFI with Secure Boot, TPM 2.0.**
- **Upgrade paths:** in-place upgrades generally keep the same edition (Home → Home). Moving Home → Pro just needs a Pro product key.

REMEMBER

If a user needs to join a domain, use BitLocker, or use gpedit, Windows Home won't do it. Upgrade to Pro.

Microsoft Windows Tools

TASK MANAGER (CTRL+SHIFT+ESC)

Tabs: **Processes** (CPU, memory, disk, network per app; end unresponsive tasks), **Performance**, **App history**, **Startup** (disable slow startup apps), **Users**, **Details**, and **Services**.

MMC SNAP-INS AND OTHER TOOLS

TOOL	SHORTCUT	USE IT TO...
Event Viewer	eventvwr.msc	Read system, application, and security logs. First stop for crashes and errors.
Disk Management	diskmgmt.msc	Create, delete, extend, format partitions. Initialize new disks. Assign drive letters.
Task Scheduler	taskschd.msc	Run programs or scripts automatically on a schedule or trigger.

TOOL	SHORTCUT	USE IT TO...
Device Manager	devmgmt.msc	Update, roll back, disable, or uninstall drivers. Spot devices with problems (yellow !).
Certificate Manager	certmgr.msc	View and manage the user's digital certificates.
Local Users and Groups	lusrmgr.msc	Manage local accounts and groups (not in Home).
Performance Monitor	perfmon.msc	Track detailed performance counters over time; build reports.
Group Policy Editor	gpedit.msc	Configure local policies (not in Home).
System Information	msinfo32.exe	Full hardware and software inventory.
Resource Monitor	resmon.exe	Detailed real-time CPU, memory, disk, and network use per process.
System Configuration	msconfig.exe	Boot options (Safe Mode, diagnostic startup), services. Startup items moved to Task Manager.
Disk Cleanup	cleanmgr.exe	Free space by removing temp files, old updates, recycle bin.
Defragment and Optimize Drives	dfrgui.exe	Defragment HDDs; TRIM SSDs. (Don't defragment SSDs.)
Registry Editor	regedit.exe	Edit the registry directly. Back up first. Mistakes can break Windows.

Windows Command-Line Tools

COMMAND	PURPOSE
cd / dir / md (mkdir) / rd (rmdir)	Change directory, list contents, make a folder, remove a folder.
copy / xcopy / robocopy	Copy files. xcopy copies folders and subfolders. robocopy is the most robust: resumes, mirrors, keeps permissions.
ipconfig	Show IP config. /all full details, /release and /renew refresh DHCP, /flushdns clears the DNS cache.
ping / tracert / pathping	Test reachability, show route hops, show per-hop loss.
nslookup	Test DNS resolution.
netstat	Show active connections and listening ports.
hostname / whoami	Show the computer name / the signed-in user.
net user / net use	Manage user accounts / map network drives (e.g., net use Z: \\server\share).

COMMAND	PURPOSE
chkdsk	Check a disk for errors. /f fixes errors, /r also locates bad sectors.
sfc /scannow	System File Checker: scans and repairs protected Windows system files.
DISM	Repairs the Windows image that sfc uses (run DISM /Online /Cleanup-Image /RestoreHealth if sfc can't fix things).
format / diskpart	Format a volume / command-line disk partitioning.
gpupdate / gpresult	Force a Group Policy refresh (/force) / show which policies applied.
shutdown	Shut down or restart (/s shutdown, /r restart, /t delay).
winver	Show the Windows version and build.
[command] /?	Help for any command.

REMEMBER

Run commands that change the system (sfc, chkdsk, diskpart) from an **elevated** prompt: right-click → Run as administrator.

Windows Settings and Control Panel

CONTROL PANEL UTILITY	USE
Internet Options	Browser proxy, security zones, cache, and connection settings.
Devices and Printers	Add and manage printers and peripherals.
Programs and Features	Uninstall or change programs, turn Windows features on or off.
Network and Sharing Center	View connections, change adapter settings, sharing options.
System	Computer name, domain/workgroup, remote settings, virtual memory, system protection.
Windows Defender Firewall	Allow or block apps and ports.
Mail	Outlook profile setup.
Sound	Playback and recording devices.
User Accounts	Manage local accounts and credentials.
Indexing Options	Control what Windows Search indexes.
File Explorer Options	Show hidden files, show file extensions , general and view options.
Power Options	Sleep, hibernate, fast startup, USB selective suspend, lid-close behavior.

CONTROL PANEL UTILITY	USE
Ease of Access / Accessibility	Narrator, magnifier, high contrast, keyboard aids.

The **Settings** app covers much of the same ground: Time and Language, Update and Security (Windows Update, recovery), Personalization, Apps, Privacy, System, Devices, Network and Internet, Gaming, and Accounts.

- **Sleep** keeps the session in RAM using a little power (fast resume). **Hibernate** saves the session to disk and powers off completely (no power used, slower resume). **Fast startup** is a partial hibernate of the kernel to speed boot.

Windows Networking Features

- **Shared resources:** share folders and printers; map drives with File Explorer or **net use**; administrative shares end in \$ (e.g., C\$, hidden).
- **Firewall:** allow apps through or create port rules. Network profiles: **Public** (strict, discovery off) vs. **Private** (trusted, discovery on).
- **Client network config:** IP, subnet mask, gateway, DNS; static vs. DHCP. **Alternate configuration** is a fallback static IP if DHCP fails.
- **Connection types:** wired, wireless, VPN, and WWAN (cellular).
- **Proxy settings:** set in Settings or Internet Options.
- **Metered connections:** limit background data use (useful on cellular).

macOS Features and Tools

FEATURE	PURPOSE
Installing apps	.dmg (disk image you mount and drag the .app from), .pkg (installer package), App Store. Uninstall by dragging the .app to Trash.
Finder	File manager (like File Explorer).
Spotlight	System-wide search (Cmd+Space).
Mission Control	View all open windows and virtual desktops (Spaces).
Dock	App launcher and switcher bar.
Keychain	Stores passwords, keys, and certificates.
iCloud	Apple's cloud sync and backup.
Gestures	Multi-touch trackpad shortcuts.
Time Machine	Built-in backup to an external or network drive.
FileVault	Full-disk encryption.

FEATURE	PURPOSE
Disk Utility	Format, partition, repair (First Aid) disks.
Terminal	Command line (zsh shell, Unix commands).
Force Quit	Cmd+Option+Esc closes frozen apps.
System Settings	Displays, Network, Printers and Scanners, Privacy and Security, Accessibility.

Linux Tools and Commands

COMMAND	PURPOSE
ls / pwd / cd	List files / print working directory / change directory.
cp / mv / rm	Copy / move or rename / delete files (rm -r for folders).
chmod	Change permissions (e.g., chmod 755 file = rwx for owner, r-x for group and others).
chown	Change file owner and group.
grep	Search text for a pattern (e.g., grep error /var/log/syslog).
find	Search for files by name, size, date, etc.
su / sudo	Switch user (often to root) / run a single command as root.
apt / dnf	Package managers: apt on Debian/Ubuntu, dnf on Red Hat/Fedora.
ip / ping / curl / dig / traceroute	Show and configure interfaces / test reachability / transfer data from URLs / query DNS / show route hops.
man	Manual page for any command.
cat / nano	Display file contents / simple text editor.
top / ps	Live process monitor / list processes.
du / df	Disk usage of files and folders / free space on file systems.
fsck / mount	Check and repair a file system / attach a file system to the directory tree.

CONFIG FILE	CONTAINS
/etc/passwd	User account info (no passwords, despite the name).
/etc/shadow	Hashed passwords (readable only by root).
/etc/hosts	Local name-to-IP mappings, checked before DNS.
/etc/fstab	File systems to mount at boot.
/etc/resolv.conf	DNS servers the system uses.

- **Linux components:** the **kernel** (core of the OS), the **bootloader** (e.g., GRUB, loads the kernel), and **systemd** (init system that starts and manages services; control them with systemctl).
- **root** is the all-powerful superuser account. Use sudo for individual admin tasks instead of logging in as root.

REMEMBER

Permission numbers: read = 4, write = 2, execute = 1. Add them per group (owner, group, others).
755 = rwx r-x r-x. 644 = rw- r-- r--.

Installing Applications

- **System requirements:** 32-bit vs. 64-bit (64-bit apps can't run on a 32-bit OS), CPU, RAM, storage, **dedicated vs. integrated graphics** and VRAM, external hardware tokens (license dongles).
- **OS requirements:** supported OS and version.
- **Distribution methods:** physical media, mountable **ISO** files, downloadable packages, image deployment.
- **Impact considerations:** on the device (performance, conflicts), the network (bandwidth during rollout), operations (training, support load), and the business (licensing cost, compliance).

Cloud-Based Productivity Tools

- **Common tools:** email systems, cloud storage with **synchronization** folders, and collaboration suites (spreadsheets, word processing, presentations, videoconferencing, instant messaging), e.g., Microsoft 365 and Google Workspace.
- **Identity synchronization:** syncing on-premises Active Directory accounts to the cloud identity service (e.g., Microsoft Entra ID), so users have one set of credentials.
- **License assignment:** admins assign product licenses to users in the cloud admin portal. No license = no access to that app.

02

Security

A+ CORE 2 · 220-1202 · DOMAIN 2 · 28% OF THE EXAM

Physical and logical controls, Windows security settings, threats and malware, and securing workstations, mobile devices, SOHO networks, and browsers. The malware removal steps are tested in a specific order.

Physical Security

CONTROL	PURPOSE
Bollards	Posts that stop vehicles from ramming buildings.
Access control vestibule	Two-door entry space where only one door opens at a time. Stops tailgating. (Formerly “mantrap.”)
Badge reader	Electronic access with ID cards.
Video surveillance	Cameras for monitoring and evidence.
Alarm systems / motion sensors	Detect and alert on intrusions.
Door locks / equipment locks	Restrict access to rooms and secure devices (cable locks for laptops).
Security guards	Human verification and deterrence.
Fences / lighting	Perimeter control and deterrence.
Magnetometers	Metal detectors at entrances.

Personnel access methods: key fobs, smart cards, mobile digital keys, physical keys, and **biometrics** (fingerprint, palm, retina, facial recognition, voice).

Logical Security Concepts

- **Principle of least privilege:** users get only the access they need.
- **Zero Trust:** never trust automatically, always verify every user and device, every time.
- **ACLs:** lists of who or what is allowed or denied access to a resource.
- **MFA methods:** email codes, hardware tokens, authenticator apps, SMS, voice call, **TOTP** (time-based one-time password), and one-time passcodes. SMS is the weakest (SIM swapping).

- **SSO (Single Sign-On)** and **SAML** (the protocol that passes authentication between an identity provider and apps).
- **JIT (Just-In-Time) access**: elevated permissions granted only for a limited time when needed.
- **PAM (Privileged Access Management)**: controls and monitors admin accounts.
- **IAM (Identity and Access Management)**: managing user identities and their permissions.
- **MDM**: enforces security on mobile devices.
- **DLP (Data Loss Prevention)**: detects and blocks sensitive data leaving the organization (e.g., credit card numbers in email).
- **Directory services**: central databases of users and resources, like Active Directory.

Windows Security Settings

DEFENDER, FIREWALL, AND ACCOUNTS

- **Microsoft Defender Antivirus**: built-in. Know how to activate/deactivate it and update its definitions.
- **Windows Defender Firewall**: turn it on/off per network profile, allow apps, and create port rules.
- **Account types**: **Administrator** (full control), **Standard user** (day-to-day use, can't change system settings), **Guest** (very limited, should be disabled), **Power User** (legacy). **Local account** vs. **Microsoft account** (cloud-synced).
- **Sign-in options**: username/password, PIN, fingerprint, facial recognition (**Windows Hello**), SSO, and passwordless sign-in.
- **UAC (User Account Control)**: prompts for consent or admin credentials before system changes. Prevents malware from silently gaining admin rights. Don't disable it.
- **Run as administrator** elevates a single program. Day-to-day work should be done as a standard user.

NTFS VS. SHARE PERMISSIONS

- **NTFS permissions** apply whether the user is local or over the network. **Share permissions** apply only when accessing over the network.
- When both apply, the **most restrictive** wins.
- **Inheritance**: files and subfolders get their parent folder's permissions by default.
- **Moving vs. copying**: moving a file within the same NTFS volume keeps its permissions. Copying, or moving to a different volume, inherits the destination folder's permissions.
- **Explicit Deny** overrides Allow.

ENCRYPTION

- **BitLocker**: full-volume encryption, uses the TPM. Not in Windows Home. Save the **recovery key**.
- **BitLocker To Go**: encrypts removable drives like USB flash drives.

- **EFS (Encrypting File System):** encrypts individual files and folders, tied to the user account. NTFS only, not in Home.

ACTIVE DIRECTORY

- **Joining a domain** puts the PC under central management.
- **Group Policy** pushes settings and updates to users and computers. **OUs (Organizational Units)** are containers used to organize objects and target policies.
- **Security groups** assign permissions to many users at once.
- **Login scripts** run at sign-in (e.g., mapping drives). **Home folders** give each user personal network storage. **Folder redirection** stores folders like Documents on the server.

Wireless Security Protocols and Authentication

- **WPA2** with AES encryption is the minimum acceptable. **WPA3** is current and stronger (SAE replaces the PSK handshake).
- **TKIP** is old and weak; **AES** is the strong, modern encryption. Always pick AES.
- **RADIUS:** central authentication for Wi-Fi and VPN (enterprise). **TACACS+:** central authentication for admin access to network devices. **Kerberos:** ticket-based authentication used by Active Directory domains.
- **MFA** for network logins adds another layer on top.

Malware Types

MALWARE	WHAT IT DOES
Virus	Attaches to files and spreads when they're run.
Boot sector virus	Infects the boot record, loading before the OS.
Trojan	Disguised as legitimate software.
Rootkit	Hides deep in the system (often the kernel) to avoid detection. Hard to remove, often requires a reimage.
Ransomware	Encrypts files and demands payment.
Spyware	Secretly monitors activity.
Keylogger	Records keystrokes to steal passwords.
Cryptominer	Hijacks CPU/GPU to mine cryptocurrency. Symptom: high resource use and heat.
Stalkerware	Spyware used to track a specific person (often on phones).
Fileless malware	Runs in memory using legit tools like PowerShell, leaving few files to scan.
Adware	Floods users with ads and pop-ups.

MALWARE	WHAT IT DOES
PUP	Potentially Unwanted Program: bundled junk software (toolbars, “optimizers”).

TOOLS AND METHODS

- **Anti-malware / antivirus and software firewalls.**
- **EDR** (Endpoint Detection and Response) monitors endpoint behavior and responds to threats. **MDR** (Managed Detection and Response) is EDR run by an outside security team. **XDR** (Extended Detection and Response) correlates data across endpoints, network, email, and cloud.
- **Email security gateway:** filters malicious mail before delivery.
- **Recovery console / WinRE and OS reinstallation** when malware can’t be removed.
- **User education**, especially anti-phishing training.

Social Engineering, Threats, and Vulnerabilities

SOCIAL ENGINEERING

- **Phishing** (email), **vishing** (voice calls), **smishing** (SMS), **QR code phishing** (“quishing,” malicious QR codes), **spear phishing** (targeted at specific people), **whaling** (targeting executives).
- **Shoulder surfing** (watching screens or keypads), **tailgating** (following someone through a secure door), **impersonation** (pretending to be IT, a vendor, etc.), **dumpster diving** (searching trash).

THREATS

THREAT	DESCRIPTION
DoS / DDoS	Overwhelm a system so it can’t serve legitimate users.
Evil twin	Fake Wi-Fi AP using a real network’s name.
Zero-day	Attack on a flaw the vendor doesn’t know about or hasn’t patched yet.
Spoofing	Faking an identity: email address, IP, MAC, caller ID.
On-path	Attacker secretly sits between two parties, intercepting traffic.
Brute force / dictionary	Guessing passwords: every combination / common words and leaked passwords.
Insider threat	Malicious or careless employees.
SQL injection	Malicious SQL entered in input fields to read or change a database.
XSS (Cross-site scripting)	Injecting malicious scripts into websites viewed by other users.
BEC (Business Email Compromise)	Impersonating executives or vendors by email to trick staff into wiring money.
Supply chain / pipeline attack	Compromising a trusted vendor or software update to reach its customers.

VULNERABILITIES

Non-compliant systems, unpatched systems, unprotected systems (no antivirus or firewall), end-of-life OSs, and BYOD devices outside IT's control.

The Malware Removal Process

Memorize these seven steps in order. A+ asks “what should the technician do NEXT?” about this list.

STEP	ACTION
1. Investigate and verify malware symptoms	Confirm it's actually malware (pop-ups, redirects, slow performance, disabled AV).
2. Quarantine infected systems	Disconnect from the network so it can't spread.
3. Disable System Restore (Windows Home)	Prevents malware hiding in restore points from coming back.
4. Remediate infected systems	Update anti-malware software, then scan and remove using techniques like Safe Mode or a preinstallation environment (WinPE) . Reimage or reinstall if needed.
5. Schedule scans and run updates	Make sure AV and the OS stay updated going forward.
6. Enable System Restore and create a restore point (Windows Home)	Creates a clean restore point.
7. Educate the end user	Explain how it happened and how to avoid it.

REMEMBER

Order matters: **quarantine before disabling System Restore, and update the anti-malware tool before scanning.**

Workstation Security Hardening

- **Data-at-rest encryption** (BitLocker, FileVault).
- **Password best practices:** length (long passphrases beat complexity), mixed character types, uniqueness, and expiration where policy requires it. Set **BIOS/UEFI passwords** too.
- **End-user practices:** screen-saver locks, log off when away (Win+L), secure critical hardware, and protect PII and passwords (no sticky notes).
- **Account management:** restrict user permissions, restrict login times, **disable the guest account**, lock accounts after failed attempts, use timeouts/screen locks, set account expiration dates (contractors).
- **Change the default administrator account's name and password.**

- **Disable AutoRun/AutoPlay** so USB drives don't run code automatically.
- **Disable unused services.**

Mobile Device Security

- **Hardening:** device encryption; screen locks (facial recognition, PIN, fingerprint, pattern, swipe; swipe is the weakest); configuration profiles.
- **Patch management:** keep the OS and apps updated.
- **Endpoint security software:** antivirus, anti-malware, content filtering.
- **Locator apps** find lost devices. **Remote wipe** erases a lost or stolen device. **Remote backup** protects data.
- **Failed login restrictions:** lockouts or wipe after too many wrong attempts.
- **Policies:** MDM, BYOD vs. corporate-owned devices, profile security requirements.

Data Destruction and Disposal

METHOD	NOTES
Shredding	Physically grinds drives into pieces.
Drilling	Holes through platters. Quick but data may be partially recoverable from fragments.
Degaussing	Strong magnet erases magnetic media. Doesn't work on SSDs or flash , and destroys HDDs.
Incinerating	Burning media completely.
Erasing / wiping	Overwrites data so it can't be recovered. Drive can be reused . Use secure erase tools for SSDs.
Low-level format	Factory-level reformat of the drive. Effective for reuse.
Standard format	Just removes the file table. Data is recoverable , so it's not secure.

- **Outsourcing:** third-party vendors provide a **certificate of destruction or recycling** as proof.
- Follow **regulatory and environmental requirements** for disposal.

SOHO Network Security

- **Router settings:** change default passwords, update firmware, IP filtering, content filtering, keep the router in a physically secure place, DHCP reservations, static WAN IP if needed, **disable UPnP** (lets apps open ports automatically), use a screened subnet for exposed devices, and secure management access (HTTPS, not from the WAN).
- **Wireless settings:** change the default SSID, optionally disable SSID broadcast (weak protection on its own), use WPA3 or WPA2-AES, configure an isolated guest network.

- **Firewall settings:** disable unused ports, use **port forwarding** only when needed (e.g., to reach a home server).

Browser Security

- Download browsers only from **trusted sources** and verify with **hashes** when possible.
- Keep the browser **patched**. Install only **trusted extensions and plug-ins**; malicious extensions can read everything you browse.
- Use a **password manager**.
- Check for **secure connections** (HTTPS with a valid certificate). Certificate warnings can mean an on-path attack.
- **Settings:** pop-up blocker, clear browsing data and cache, private browsing mode, sign-in/browser data sync, ad blockers, proxy settings, and secure DNS (DNS over HTTPS).

03

Software Troubleshooting

A+ CORE 2 · 220-1202 · DOMAIN 3 · 23% OF THE EXAM

Like Core 1's troubleshooting domain, this is all about matching symptoms to causes and choosing the least disruptive fix first.

Windows OS Issues

SYMPTOM	COMMON CAUSES AND FIXES
Blue screen of death (BSOD)	Bad drivers, failing RAM, overheating, corrupted system files. Note the stop code, check Event Viewer, roll back recent drivers or updates, run memory diagnostics and sfc.
Degraded performance	Too many startup apps, not enough RAM, full disk, malware. Check Task Manager, disable startup items, add resources.
Boot issues / "No OS found"	Wrong boot order, damaged boot files or boot sector, failed drive. Use Startup Repair in WinRE, check boot order.
Frequent shutdowns	Overheating, power issues, failing hardware.
Services not starting	Dependency failure, wrong account permissions, corrupted files. Check services.msc and Event Viewer.
Applications crashing	Update, repair, or reinstall the app. Verify system requirements.
Low memory warnings	Too many programs open, memory leak. Close apps, add RAM, adjust virtual memory.
USB controller resource warnings	Too many devices on one controller. Move devices to a different port or controller.
System instability	Drivers, hardware, corrupted OS files. sfc / DISM, roll back updates.
Slow profile load	Large roaming profile, network delays, corrupted profile. Rebuild the profile.
Time drift	Bad CMOS battery or NTP sync issues. Can break domain logins (Kerberos needs accurate time).

STANDARD FIXES, ROUGHLY LEAST TO MOST DISRUPTIVE

Reboot → restart services → uninstall/reinstall/update apps → add resources → verify requirements → **sfc** → **repair Windows** → **restore** (System Restore point) → **roll back updates** →

rebuild the Windows profile → reimage.

REMEMBER

Pick the least invasive fix that solves the problem. Reimaging is almost never the first answer.

Mobile OS and Application Issues

SYMPTOM	COMMON FIXES
App fails to launch, close, or update	Force-stop the app, clear its cache, update or reinstall it, check storage space.
App slow to respond	Close background apps, restart the device, free up storage.
OS fails to update	Low storage, low battery, poor connection, unsupported device.
Poor battery life	Background apps, high brightness, aging battery, malware.
Random reboots	Overheating, faulty app, failing battery, OS bug.
Bluetooth / Wi-Fi / NFC issues	Toggle the radio, forget and re-pair, restart, reset network settings.
Screen doesn't autorotate	Rotation lock is on, or the app doesn't support rotation.

Mobile OS and Application Security Issues

SECURITY CONCERNS

- **APK sideloading:** installing Android apps from outside the Play Store. Bypasses store security checks.
- **Developer mode** enabled on a regular user's device.
- **Root access / jailbreak:** removes the OS's built-in security restrictions. Unauthorized apps can then do anything.
- **Bootleg / malicious apps** and **app spoofing** (fake apps imitating real ones).

SYMPTOMS

High network traffic, sluggish response, **data-usage limit notifications**, limited or no internet, lots of ads, fake security warnings, unexpected app behavior, and leaked personal files or data.

PC Security Issues

SYMPTOM	WHAT IT SUGGESTS
Unable to access the network	Malware may change network settings or proxy to block security sites.
Desktop alerts / false antivirus alerts	Fake (“scareware”) security warnings pushing you to install something.
Altered system or personal files	Missing, renamed, or inaccessible files. Possibly ransomware.
Unwanted OS notifications	Malicious browser notification permissions or adware.
OS update failures	Malware often blocks updates to protect itself.
Random or frequent pop-ups	Adware or a malicious extension.
Certificate warnings	On-path attack, wrong system time, or an actually invalid site.
Browser redirection	Hijacked search engine, malicious extension, modified hosts file, or DNS changes.
Degraded browser performance	Too many or malicious extensions, adware, cryptominer.

REMEMBER

Certificate warnings on *every* site? Check the system date and time first. A wrong clock makes valid certificates look invalid.

04

Operational Procedures

A+ CORE 2 · 220-1202 · DOMAIN 4 · 21% OF THE EXAM

This covers how professional IT shops actually work: tickets, documentation, change control, backups, safety, legal issues, communication, scripting, remote access, and AI basics.

Documentation and Support Systems

- **Ticketing systems** track: user info, device info, a description of the problem, categories, **severity**, **escalation levels**, and clear written communication (problem description, progress notes, resolution).
- **Asset management**: inventory lists, a **CMDB** (Configuration Management Database), asset tags and IDs, the procurement life cycle, warranty and licensing info, and assigned users.
- **Document types**: incident reports, **SOPs** (Standard Operating Procedures), custom software install procedures, **new-user setup checklists** (onboarding), and **end-user termination checklists** (offboarding: disable accounts, collect equipment).
- **Knowledge base**: articles documenting known problems and fixes so the team doesn't solve the same issue twice.
- **SLA (Service-Level Agreement)**: defines expected response and resolution times.

Change Management

- **Documented business processes**: a **rollback plan** (how to undo the change), a **backup plan**, **sandbox testing** (test in an isolated environment first), and a **responsible staff member**.
- **Change request contents**: purpose of the change, scope, change type, date and time (maintenance window), affected systems and impact, risk analysis and risk level.
- **Approval**: a **change advisory board (CAB)** reviews and approves.
- **After approval**: implementation, peer review, and **end-user acceptance**.

REMEMBER

Never make an unapproved change on a production system, even if it's "quick." Change management questions usually reward following the process.

Backup and Recovery

BACKUP TYPE	WHAT IT BACKS UP	RESTORE NEEDS	ARCHIVE BIT
Full	Everything	Just the last full	Cleared
Incremental	Changes since the last backup of any type	Last full + every incremental since	Cleared
Differential	Changes since the last full	Last full + the latest differential	Not cleared
Synthetic full	Built by combining the last full with later incrementals (no new full read from the source)	Just the synthetic full	n/a

REMEMBER

Incremental = fastest backups, slowest restore. Differential = slower backups (grows each day), faster restore.

- **Backup testing:** regularly test restores. An untested backup isn't a backup.
- **Onsite vs. offsite:** onsite restores faster; offsite (or cloud) survives fires, floods, and theft.
- **GFS (Grandfather-Father-Son) rotation:** daily (son), weekly (father), monthly (grandfather) backups.
- **3-2-1 rule:** 3 copies of data, on 2 different media types, with 1 copy offsite.

Safety Procedures

- **ESD (electrostatic discharge):** use an **ESD wrist strap** and **ESD mat**, store parts in antistatic bags, and touch grounded metal. ESD can damage parts without you feeling it.
- **Electrical safety:** equipment grounding, PPE. **Disconnect power before working on a PC.** Never open a power supply or CRT monitor (they hold dangerous charges even when unplugged).
- **Lifting:** lift with your legs, keep your back straight, get help with heavy items.
- **Electrical fires:** use a **Class C** extinguisher (CO2 or other non-conductive agent). **Never water.**
- **Personal protection:** safety goggles, air filtration masks (toner, dust).
- Follow **government regulations** (e.g., OSHA) for workplace safety.

Environmental Impacts and Controls

- **SDS / MSDS (Safety Data Sheets):** explain how to handle and dispose of hazardous materials. Check them for **batteries, toner**, and other chemicals.
- **Disposal:** batteries and electronics contain toxic materials. Recycle them according to local regulations, never in the trash. Toner goes back to recycling programs.
- **Temperature, humidity, and ventilation:** keep equipment cool, and avoid low humidity (static) and high humidity (condensation). Place equipment with good airflow. Clean dust with

compressed air or an electronics-safe vacuum.

- **Power protection:** **surge suppressors** guard against spikes; a **UPS / battery backup** handles **under-voltage events** (brownouts, sags) and short power failures so you can shut down safely.

Prohibited Content, Privacy, Licensing, and Policy

INCIDENT RESPONSE

- **Chain of custody:** document everyone who handles evidence, when, and why, so it holds up legally.
- **Inform management or law enforcement** as appropriate (e.g., finding illegal content).
- **Preserve data integrity:** make a forensic copy of the drive instead of working on the original.
- **Document** everything.

LICENSING

- **Valid licenses:** using unlicensed software is illegal.
- **Perpetual license** (buy once, use forever) vs. subscription. **Personal-use** vs. **corporate-use** licenses.
- **EULA** (End User License Agreement) and **DRM** (Digital Rights Management) control how software and content can be used.
- **Open source:** source code is publicly available; licenses vary in what they allow.

AGREEMENTS AND REGULATED DATA

- **NDA (Non-Disclosure Agreement):** legal promise not to share confidential information. **AUP (Acceptable Use Policy):** what users may and may not do with company systems.
- **Regulated data:** credit card payment information (PCI DSS), government-issued IDs, **PII** (Personally Identifiable Information), healthcare data (HIPAA). Know your **data retention requirements**.

Professionalism and Communication

- Use proper language and **avoid jargon** with users.
- Maintain a positive attitude and project confidence.
- **Actively listen**, take notes, and **don't interrupt**. Clarify what the user said by restating it.
- Be culturally sensitive and use appropriate titles.
- Be on time. If you'll be late, contact the customer.
- **Avoid distractions:** no personal calls, texting, or social media on the job.
- **Difficult customers:** don't argue or be defensive, don't dismiss their problem, don't post about it on social media.

- **Set and meet expectations:** give a timeline, offer repair or replace options, provide documentation, and follow up.
- **Confidential materials:** don't look at, copy, or discuss customers' private information, documents, or screens.

Scripting Basics

EXTENSION	LANGUAGE	PLATFORM
.bat	Batch file	Windows (Command Prompt)
.ps1	PowerShell	Windows (and cross-platform PowerShell 7)
.vbs	VBScript	Windows (legacy, being phased out)
.sh	Shell script (Bash/zsh)	Linux, macOS
.js	JavaScript	Web browsers, Node.js
.py	Python	Cross-platform

- **Use cases:** basic automation, restarting machines, remapping network drives, installing applications, automated backups, gathering information or data, and initiating updates.
- **Risks:** unintentionally introducing malware (running untrusted scripts), inadvertently changing system settings, and browser or system crashes from mishandling resources (e.g., infinite loops).

Remote Access Technologies

METHOD	NOTES
RDP	Windows Remote Desktop, TCP 3389. Never expose directly to the internet; use a VPN or gateway.
VPN	Encrypted tunnel into the private network.
VNC	Cross-platform screen sharing. Older versions weakly encrypted.
SSH	Secure command line, TCP 22. Use key-based authentication.
RMM	Remote Monitoring and Management: tools MSPs use to monitor, patch, and support many clients. A big target for attackers.
SPICE	Protocol for accessing virtual machine desktops.
WinRM	Windows Remote Management: run PowerShell commands on remote machines.
Third-party tools	Screen-sharing, videoconferencing, file transfer software, desktop management tools.

Security considerations: require MFA, use encryption, limit who can connect, and watch for scammers who use legitimate remote tools in tech-support scams.

Artificial Intelligence Basics

- **Application integration:** AI is built into productivity apps, help desks, search, and coding tools.
- **Policy:** organizations should have an **appropriate-use policy** for AI. Submitting AI-generated work as your own can count as **plagiarism**.
- **Limitations:** AI can show **bias**, make up false information (**hallucinations**), and be inaccurate. Always verify its output.
- **Private vs. public AI:** public AI tools may store or train on what you enter. **Never paste sensitive, customer, or company data into public AI tools** unless the organization has approved it. Consider data security, data source, and data privacy.

Practice Questions

33 QUESTIONS • ANSWER KEY FOLLOWS

- 01 A user can't copy a 7 GB video file to a USB flash drive that has 30 GB free. What is the MOST likely cause?**
- A The drive is formatted NTFS
 - B The drive is formatted FAT32
 - C The drive is write-protected by BitLocker
 - D The file is corrupted
- 02 Which Windows edition is the minimum needed to join an Active Directory domain?**
- A Home
 - B Pro
 - C Enterprise
 - D Pro for Workstations
- 03 Which is a requirement for installing Windows 11?**
- A TPM 2.0
 - B 2 GB RAM
 - C Legacy BIOS
 - D MBR partitioning
- 04 A technician needs to roll back a recently updated network adapter driver. Which tool should be used?**
- A Disk Management
 - B Device Manager
 - C Event Viewer
 - D Task Scheduler
- 05 Which command repairs protected Windows system files?**
- A `chkdsk /f`
 - B `sfc /scannow`
 - C `gpupdate /force`
 - D `ipconfig /renew`
- 06 A policy change was made on the domain controller, and a user's PC needs it right away. Which command should be run on the PC?**
- A `gpresult`
 - B `gpupdate /force`
 - C `net use`
 - D `winver`

- 07 Which command copies a folder structure while preserving permissions and resuming after interruptions?**
- A copy
 - B xcopy
 - C robocopy
 - D move
- 08 On a Linux system, which command installs a package on Ubuntu?**
- A dnf install
 - B apt install
 - C yum get
 - D pkg add
- 09 A Linux file needs owner read/write/execute and read/execute for everyone else. Which command is correct?**
- A chmod 644
 - B chmod 755
 - C chmod 777
 - D chown 755
- 10 Which macOS feature provides full-disk encryption?**
- A Time Machine
 - B Keychain
 - C FileVault
 - D Spotlight
- 11 A user accesses a shared folder over the network. The share permission is Full Control, and the NTFS permission is Read. What is the effective permission?**
- A Full Control
 - B Read
 - C Modify
 - D No access
- 12 Which Windows feature encrypts a USB flash drive?**
- A EFS
 - B BitLocker To Go
 - C UAC
 - D Windows Hello

- 13 Which feature prompts the user for permission before a program makes system-level changes?**
- A UAC
 - B Defender Firewall
 - C EFS
 - D Group Policy
- 14 An employee receives a text message with a link claiming to be from their bank. What attack is this?**
- A Vishing
 - B Whaling
 - C Smishing
 - D Tailgating
- 15 The CFO receives an email that appears to come from the CEO, urgently requesting a wire transfer to a new vendor. What is this?**
- A Business email compromise
 - B Evil twin
 - C SQL injection
 - D Shoulder surfing
- 16 A technician has verified malware symptoms on a Windows Home PC. What is the NEXT step?**
- A Disable System Restore
 - B Quarantine the infected system
 - C Educate the user
 - D Run a scan
- 17 During remediation, what should be done BEFORE scanning for malware?**
- A Educate the end user
 - B Update the anti-malware software
 - C Create a restore point
 - D Re-enable System Restore
- 18 Which data destruction method is NOT effective on SSDs?**
- A Shredding
 - B Incinerating
 - C Degaussing
 - D Secure erase

- 19 A company wants to reuse old hard drives in other departments. Which method is BEST?**
- A Standard format
 - B Drilling
 - C Wiping / overwriting
 - D Degaussing
- 20 Which SOHO router feature should be disabled because it lets devices open ports automatically?**
- A DHCP
 - B UPnP
 - C NAT
 - D WPA3
- 21 A user gets certificate warnings on every website, including major ones. What should be checked FIRST?**
- A The system date and time
 - B The printer driver
 - C The BIOS password
 - D The screen resolution
- 22 A user's browser keeps redirecting searches to an unfamiliar search engine. What is the MOST likely cause?**
- A Bad network cable
 - B Malicious browser extension or hijacker
 - C Expired DHCP lease
 - D Failing hard drive
- 23 An Android user installed an app from a website instead of the Play Store, and now sees constant ads. What practice caused this risk?**
- A Jailbreaking
 - B APK sideloading
 - C Developer mode
 - D Airplane mode
- 24 A user's Windows profile loads very slowly and settings keep resetting. Other users on the PC are fine. What is the BEST fix?**
- A Reimage the PC
 - B Rebuild the user's profile
 - C Replace the RAM
 - D Reinstall all apps

- 25 A phone's screen won't rotate when turned sideways in any app. What should be checked FIRST?**
- A Rotation lock setting
 - B Battery health
 - C Factory reset
 - D Replace the digitizer
- 26 Which backup type backs up only changes since the last full backup?**
- A Full
 - B Incremental
 - C Differential
 - D Synthetic
- 27 What does the 3-2-1 backup rule recommend?**
- A 3 backups per day, 2 per week, 1 per month
 - B 3 copies, 2 media types, 1 offsite
 - C 3 servers, 2 sites, 1 cloud
 - D 3 full, 2 incremental, 1 differential
- 28 Before implementing a change to a production server, where should it be tested?**
- A On the CEO's PC
 - B In a sandbox
 - C Directly in production after hours
 - D It doesn't need testing
- 29 Which fire extinguisher class is used for electrical fires?**
- A Class A
 - B Class B
 - C Class C
 - D Water
- 30 A technician finds illegal content on a customer's computer during a repair. What should the technician do?**
- A Delete it
 - B Follow incident response: preserve evidence, maintain chain of custody, and report to management
 - C Ignore it
 - D Post about it

- 31 Which file extension is a PowerShell script?**
- A .bat
 - B .vbs
 - C .ps1
 - D .sh
- 32 An employee wants to paste a customer database into a free public AI chatbot to summarize it. What is the BEST response?**
- A Allow it, AI is accurate
 - B Don't. Public AI tools may store the data, which risks customer privacy
 - C Allow it if the file is small
 - D Allow it only after hours
- 33 A customer is angry and raising their voice about a repeated problem. What should the technician do?**
- A Argue back to establish authority
 - B Stay calm, listen without interrupting, and restate the problem
 - C Leave immediately
 - D Tell them it's their fault

Answer Key

#	ANSWER	WHY
01	B	FAT32 has a 4 GB maximum file size. Reformat as exFAT. (Domain 1)
02	B	Home can't join a domain; Pro is the minimum. (Domain 1)
03	A	Windows 11 requires TPM 2.0, UEFI with Secure Boot, 4 GB RAM, and 64 GB storage. (Domain 1)
04	B	Device Manager can roll back drivers. (Domain 1)
05	B	System File Checker repairs system files. (Domain 1)
06	B	gpupdate /force refreshes Group Policy immediately. (Domain 1)
07	C	robocopy is the robust copy tool. (Domain 1)
08	B	apt is the Debian/Ubuntu package manager. dnf is for Red Hat/Fedora. (Domain 1)
09	B	7 = rwx, 5 = r-x. (Domain 1)
10	C	FileVault encrypts the startup disk. (Domain 1)
11	B	When share and NTFS permissions combine, the most restrictive wins. (Domain 2)
12	B	BitLocker To Go encrypts removable drives. (Domain 2)
13	A	User Account Control. (Domain 2)
14	C	Smishing is SMS phishing. (Domain 2)
15	A	BEC impersonates executives or vendors to trick staff into sending money. (Domain 2)
16	B	Step 2 is quarantine. Disabling System Restore is step 3. (Domain 2)
17	B	Update definitions first so the scan detects current threats. (Domain 2)
18	C	SSDs don't store data magnetically, so degaussing does nothing. (Domain 2)
19	C	Wiping makes data unrecoverable while leaving the drive usable. A standard format leaves data recoverable. (Domain 2)
20	B	UPnP can be abused by malware to open ports. (Domain 2)
21	A	A wrong clock makes valid certificates appear invalid. (Domain 3)
22	B	Redirection is a classic sign of browser hijacking. (Domain 3)
23	B	Sideloaded APKs bypasses Play Store security checks. (Domain 3)
24	B	The problem is specific to one profile, so rebuild it. (Domain 3)
25	A	Start with the simplest cause. (Domain 3)
26	C	Differential = since last full. Incremental = since last backup of any type. (Domain 4)

#	ANSWER	WHY
27	B	Three copies, two media types, one offsite. (Domain 4)
28	B	Sandbox testing is part of change management. (Domain 4)
29	C	Class C covers energized electrical equipment. (Domain 4)
30	B	Preserve evidence and follow proper reporting procedures. (Domain 4)
31	C	.ps1 is PowerShell. (Domain 4)
32	B	Sensitive data shouldn't go into unapproved public AI tools. (Domain 4)
33	B	Professionalism means listening, not getting defensive. (Domain 4)